(Please write your Enrollment Number) Enrollment No._____

END-TERM EXAMINATION
CBCS)(SUBJECTIVE TYPE)(OFFLINE)
Course Name:<B.Tech. (CSEAI)>, Semester:<6>
(May, 2026)

Subject Code: BAI 302	Subject: Natural Language Processing
Time : 3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	**Answer the following**	(2.5*8=20)
	a) How does ambiguity in word forms affect the efficiency of Morphological analysis in NLP?	
	b) If cosine similarity between two documents is high but topics are different, analyze possible reasons and suggest improvements.	
	c) Identify the morphological type (Noun phrase, Verb Phrase, Adjective Phrase) of following sentence segments 1. Important to Bill 2. Looked up the tree	
	d) Discuss Skip-gram model in details with suitable example.	
	e) How does NLTK support deep learning workflows?	
	f) What is smoothing and reordering in Statistical Machine Translation?	
	g) Explain the role of deep learning models in NLP?	
	h) Explain why RNN is suitable for sequence data?	
	UNIT I	
Q2	**Attempt any two parts**	(5*2=10)
	a) Explain the levels of linguistic analysis in NLP: phonology, morphology, syntax, and semantics with examples.	
	b) Discuss syntactic parsing and its role in sentence understanding.	
	c) Construct the parse tree for the sentence. "The girl plucked the flower with a long stick." Discuss the ambiguity arises from the parse tree.	
	UNIT II	
Q3	**Attempt any two parts**	(5*2=10)
	a) Assume the sentence below. Create TF-IDF vectors using Term- document matrix S1: I do not like Vanila Cake S2: : I do not like Vanila Icecream	
	b) Explain what is Word2vec. Word2vec maps "king- man + woman=queen". Explain mathematically and conceptually how such relationships are learned.	
	c) Explain how GloVe captures multiple contexts and its limitation in this case.	
	UNIT III	
Q4	**Attempt any two parts**	(5*2=10)
	a) Compare rule-based, statistical, and neural machine translation by explaining how each handles the sentence: "He gave her a ring".	
	b) Explain the working of the Anusaarka system and how it differs from fully automatic machine translation.	
	c) Explain the architecture of a Neural Machine Translation (NMT) system and explain how attention mechanism improves translation quality compared to basic encoder-decoder models.	

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	UNIT IV	
Q5	**Attempt any two parts**	**(5*2=10)**
	a) Compare RNN, LSTM, and GRU in terms of architecture and performance in NLP task.	
	b) Discuss the role of pre-trained models (like embeddings) in improving NLP performance. Why does deep learning require large amounts of data in NLP tasks?	
	c) A sentiment analysis model fails on long reviews. Explain why basic RNN fails and how LSTM and GRU improves performance. Also, explain how CNN can be used in NLP, especially for sentence classification.	

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